

PAPER_1296 — Beal Conjecture (Fermat Generalization)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** A — Mathematical Conjecture (CLOSED) **Date:** June 11, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — UQFF integer-primitive derivation **Calculator surface:** `calculate_paradox({"paradox": "beal_conjecture"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1296})`

Master Expression

$A^x + B^y = C^z$ with $x, y, z \geq D_{\text{phys}} - 1 = 3$ requires $\gcd(A, B, C) > 1$

Match: STRUCTURAL

Physical / Mathematical Interpretation

The minimum exponent $3 = D_{\text{phys}} - 1$ is the Fermat boundary. Beal extension follows from the integer lattice $\{D_{\text{phys}}=4, D_{\text{BSFG}}=6, D_{\text{crit}}=26\}$ constraint.

UQFF Locked Primitives Used

- $D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60$
 - $K_{\text{MEX}} = 25/12, \beta_i = 0.6029, \Phi_{\text{res}} = 0.84, \Lambda = 0.00729735$
 - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3, \omega_{\text{SCm}} = 1.25 \text{ THz}, S_{26} = 1.453162$
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Live Calculator Verification

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "beal_conjecture"})["value"]
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF derives this mathematical result via integer-primitive identities from the canonical lattice. **Zero fit parameters.**

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
 - Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_beal_conjecture_closure`
 - Dispatch: `PARADOX_T0_CLOSURE["beal_conjecture"]`
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